Contest Problem Set 12127

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Number Sense 12127

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Name _____

Grade _____

School _____

1. $221 - 20 =$ _____.
2. $7 + 7 \times 6 =$ _____.
3. $724 - 379 =$ _____.
4. The hundreds digit of 951413 is _____.
5. $437 + 371 =$ _____.
6. $34 + 46 + 22 + 29 =$ _____.
7. $20 \times 37 =$ _____.
8. The remainder of $64 \div 3$ is _____.
9. $84 \div 7 =$ _____.
10. (estimate) $362 + 543 + 905 =$ _____.
11. $11 \times 87 =$ _____.
12. $624 - 285 - 217 =$ _____.
13. $14^2 =$ _____.
14. The remainder of $476 \div 9$ is _____.
15. $45 \times 26 =$ _____.
16. $5 \times 13 \times 16 =$ _____.
17. $319 + 145 + 116 =$ _____.
18. *CMLXXXVII* in Arabic numerals is _____.
19. $34 \times 46 =$ _____.
20. (estimate) $441 \times 499 =$ _____.
21. If 4 quarts is equal to 1 gallon, then $\frac{1}{2}$ gallon is equal to _____ quarts.
22. $86 \times 25 =$ _____.
23. The GCD of 66 and 42 is _____.
24. $28 \times 35 \div 49 =$ _____.
25. $24^2 =$ _____.
26. $\frac{1}{8} + \frac{1}{12} + \frac{1}{24} =$ _____ (fraction).
27. $18 + 23 + 28 + 33 + 38 + 43 =$ _____.
28. The smaller of $\frac{3}{19}$ and $\frac{2}{13}$ is _____ (fraction).
29. $612 \div 18 =$ _____.
30. (estimate) $49 \times 62 \times 39 =$ _____.
31. $85^2 =$ _____.
32. The number of multiples of 7 between 32 and 123 is _____.
33. $99 \times 43 =$ _____.
34. The area of a square with perimeter 60 is ____.
35. 5400 seconds is _____ hours (decimal).
36. $\sqrt{784} =$ _____.
37. The LCM of 66 and 42 is _____.
38. $76 \times 74 =$ _____.
39. 75% of 56 is _____.
40. (estimate) $142857 \times 43 =$ _____.

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41. $101 \times 293 =$ _____.

42. $16 \times 6 + 8 \times 8 =$ _____.

43. The side length of a regular octagon with perimeter 248 is _____.

44. $68^2 - 42^2 =$ _____.

45. The number of terms in the arithmetic sequence $6, 18, 30, \ldots, 186$ is _____.

46. $9^4 =$ _____.

47. $23 \times \frac{23}{27} =$ _____ (mixed number).

48. The remainder of $5328 \div 11$ is _____.

49. $98 \times 18 =$ _____.

50. (estimate) $167 \times 852 =$ _____.

51. $118 \times 105 =$ _____.

52. The number of diagonals in a regular hexagon is _____.

53. $333 \times 41 =$ _____.

54. The median of the list $1, 2, 4, 8, 16, 32$ is _____.

55. $\sqrt{2209} =$ _____.

56. 313_4 in base 2 is _____ $_2$

57. $84 \times 94 =$ _____.

58. The sum of the terms of the arithmetic sequence $9, 19, 29, \ldots, 199$ is _____.

59. $44\frac{4}{9}\%$ of 162 is _____.

60. (estimate) $\sqrt[3]{797979} =$ _____.

61. $625 \times 88 =$ _____.

62. $0.1\overline{5} =$ _____ (fraction);

63. Two cards are selected without replacement from a standard deck of cards. The probability both are kings is _____ (fraction).

64. $16\frac{4}{9} \times 2\frac{4}{9} =$ _____ (mixed number).

65. The perimeter of an equilateral triangle with height $15\sqrt{3}$ is _____.

66. $8 \times 9 \times 12 \times 13 =$ _____.

67. In a right triangle one angle measures 49° and the measure of the other acute angle is ____ $^\circ$.

68. If $v = 21$, then $v^2 + 16v + 64 =$ _____.

69. The number 245 written in base 9 is _____ $_9$.

70. (estimate) $9.5^2 \times 8.5^2 =$ _____.

71. $\sqrt{48} \times \sqrt{108} =$ _____.

72. The height of a right circular cylinder with radius 3 and volume 144π is _____.

73. $975 \times 976 =$ _____.

74. $705^2 =$ _____.

75. The perimeter of a right triangle with legs of length 10 and 24 is _____.

76. $\frac{3}{7}$ of 51 is $\frac{17}{14}$ of _____.

77. $1024^{0.7} =$ _____.

78. $46 \times 75 \times 18 =$ _____.

79. The ninth term in the geometric sequence $\frac{4}{81}, \frac{4}{27}, \frac{4}{9}, \ldots$ is _____.

80. (estimate) $\sqrt{20 \times 21 \times 22 \times 23 \times 24} =$ _____.

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Select E only if you cannot determine a uniquely correct answer between A, B, C, and D.

1.	A	B	C	D	E	11.	A	B	C	D	E	21.	A	B	C	D	E
2.	A	B	C	D	E	12.	A	B	C	D	E	22.	A	B	C	D	E
3.	A	B	C	D	E	13.	A	B	C	D	E	23.	A	B	C	D	E
4.	A	B	C	D	E	14.	A	B	C	D	E	24.	A	B	C	D	E
5.	A	B	C	D	E	15.	A	B	C	D	E	25.	A	B	C	D	E
6.	A	B	C	D	E	16.	A	B	C	D	E	26.	A	B	C	D	E
7.	A	B	C	D	E	17.	A	B	C	D	E	27.	A	B	C	D	E
8.	A	B	C	D	E	18.	A	B	C	D	E	28.	A	B	C	D	E
9.	A	B	C	D	E	19.	A	B	C	D	E	29.	A	B	C	D	E
10.	A	B	C	D	E	20.	A	B	C	D	E	30.	A	B	C	D	E

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1. Benton calculated the value of 12×14, and Carley calculated the value of 24×6. What is the value of Benton's result minus Carley's result?

(A) 24 (B) 36 (C) 12 (D) 30 (E) Other

2. On Sunday, Jaclyn bought two dozen oranges at the store. She used five oranges each day to make her morning orange juice on Monday, Tuesday, Wednesday, and Thursday. How many oranges did she have left over to make her morning orange juice on Friday?

(A) 3 (B) 7 (C) 6 (D) 4 (E) Other

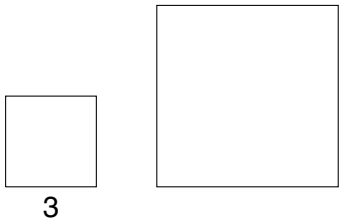
3. What is the thousands digit of 567×999?

(A) 7 (B) 4 (C) 8 (D) 6 (E) Other

4. Quarters are coins worth 25 cents and dimes are coins worth 10 cents. Tamir has fifteen quarters and a number of dimes. The total value of his quarters and dimes is 485 cents. How many dimes does Tamir have?

(A) 14 (B) 11 (C) 15 (D) 12 (E) Other

5. The smaller square below has a side of length 3. The larger square has a side twice as long as the side of the smaller square. What is the total area of the two squares?



(A) 54 (B) 36 (C) 27 (D) 45 (E) Other

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6. What is the greatest three-digit positive whole number that is 1 more than a multiple of 27?

(A) 982 (B) 999 (C) 972 (D) 991 (E) Other

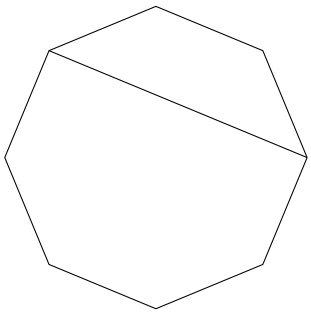
7. On a hiking trip, Jesse hiked at an average speed of three miles per hour for the first two hours. Then the trail became steeper, so Jesse's average speed for the final hour of the hike was only two miles per hour. How long was Jesse's hike, in miles?

(A) 10 (B) 6 (C) 8 (D) 9 (E) Other

8. How many minutes pass from noon on March 1st until noon on March 16th?

(A) 21040 (B) 21600 (C) 20160 (D) 23040 (E) Other

9. A diagonal is a line connecting one corner of a shape to another corner of the same shape, but a side of the shape is not a diagonal. An octagon with one diagonal drawn is shown below. How many total diagonals does the octagon have?



(A) 12 (B) 16 (C) 20 (D) 18 (E) Other

10. What whole number is nearest in value to $\frac{1}{1}+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\frac{1}{6}+\frac{1}{7}+\frac{1}{8}+\frac{1}{9}$?

(A) 2 (B) 4 (C) 3 (D) 5 (E) Other

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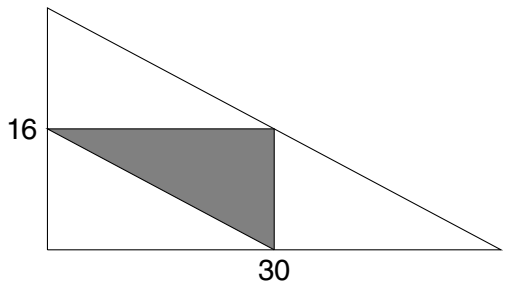
11. There are two seats in the front of a car, one of which is the driver's seat, and two seats in the back of the car. If Amelia and Daniel are the only people that can sit in the driver's seat, then how many different ways can Amelia, Bella, Charles, and Daniel be seated in the car?

(A) 24 (B) 8 (C) 4 (D) 12 (E) Other

12. What is $16\frac{2}{3}\%$ less than 25% more than 288?

(A) 336 (B) 324 (C) 320 (D) 300 (E) Other

13. The right triangle shown below has legs of length 16 and 30. The smaller gray triangle is formed by connecting the midpoints of the right triangle. What is the total area of the three regions of the right triangle that are not gray?



(A) 120 (B) 60 (C) 360 (D) 240 (E) Other

14. The nineteen students in Mrs. Copeland's fourth grade class all ranked chocolate, vanilla, and strawberry ice cream in order from most favorite flavor to least favorite flavor. Fifteen students listed strawberry as either their second or least favorite flavor, and eleven students listed chocolate as either their second or least favorite flavor. How many students listed vanilla as their most favorite flavor?

(A) 8 (B) 7 (C) 5 (D) 6 (E) Other

15. How many positive whole numbers evenly divide both 360 and 108?

(A) 9 (B) 6 (C) 10 (D) 12 (E) Other

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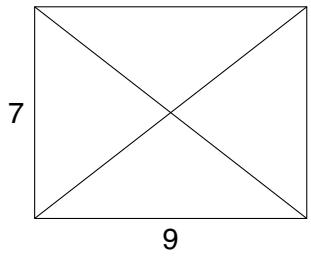
16. Evelyn is placing chocolate chip, peanut butter, and sugar cookies on small serving trays for her party, and she wants to have at least one of every type of cookie on a tray. How many different ways can Evelyn select six cookies to place on a tray? For example, one such possibility is two chocolate chip cookies, one peanut butter cookie, and three sugar cookies.

(A) 9 (B) 12 (C) 6 (D) 8 (E) Other

17. A rectangle has length 9 and width 7. The sum of the lengths of the diagonals of the rectangle is nearest to what whole number?



(A) 25 (B) 23 (C) 22 (D) 24 (E) Other

18. On a recent mathleague.org Sprint round, the average score of 135 competitors at at one contest site was 52, and the average competitor score of 189 competitors at a second contest site was 64. What was the average competitor score for all competitors at both contests sites?

(A) 59 (B) 60 (C) 58 (D) 59.5 (E) Other

19. A certain 2022-digit number is the digit 1 followed by 2021 instances of the digit 0. When 2021 is subtracted from this number, what is the sum of the digits of the result?

(A) 18194 (B) 18189 (C) 18176 (D) 18181 (E) Other

20. A row of seats in an auditorium has 50 seats, and students are sitting in some of the seats. When Janice was choosing a seat in the row, she noticed that no matter which seat she chose, she would have to sit next to another student. What is the fewest number of students that could have been sitting in the row when Janice was choosing her seat?

(A) 20 (B) 17 (C) 25 (D) 24 (E) Other

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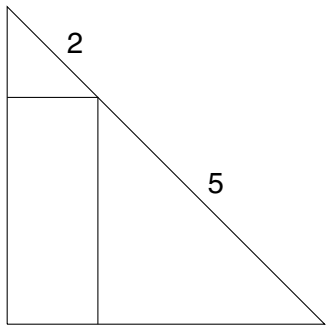
21. If $a\heartsuit b = a^2 + 2ab + b^2$, then what is the value of $(91\heartsuit 8) - (41\heartsuit 8)$?

(A) 7550 (B) 7450 (C) 7400 (D) 7350 (E) Other

22. A point on the hypotenuse of an isosceles right triangle divides the hypotenuse into segments of length 2 and 5. Segments meeting at that point and parallel to the sides of the triangle partition the triangle into two triangles and a rectangle, as shown. What is the area of the rectangle?



(A) $5\sqrt{2}$ (B) 10 (C) 7 (D) 5 (E) Other

23. If $5^{2-x} = 12.5$, then what is the value of 125^{1+x}?

(A) 15625 (B) 625 (C) 1000 (D) 500 (E) Other

24. Of the 154 students eating lunch in the cafeteria, 68 were wearing shorts and the rest were wearing jeans. After lunch, the students exited the cafeteria as pairs. Among these 77 pairs of students, there were 29 pairs of students where both students were wearing shorts. Among how many pairs of students did both students wear jeans?

(A) 39 (B) 41 (C) 42 (D) 43 (E) Other

25. Two cards are dealt from a standard deck of cards. The probability that exactly one of the cards is a jack, queen, or king is the common fraction $\frac{a}{b}$. What is the sum of a and b?

(A) 305 (B) 301 (C) 303 (D) 307 (E) Other

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26. Evan summed all of the positive whole numbers that divided 576 and which were multiples of 3, and then he divided that result by the sum of all of the positive whole numbers that divided 576 and which were not multiples of 3. What was Evan's result?

(A) 9 (B) 11 (C) 12 (D) 10 (E) Other

27. Working together and at the same rate, three workers can assemble five custom Volta electric cars in thirty days. Five workers begin assembling seven custom Volta electric cars, but after five days of work, their boss Mr. Muss removes two of the workers so that they can work on another another project. For the final seven days of assembly, those two workers return. For the start of assembly until the work is complete, how long, in days, does it take the workers to assemble the seven custom Volta electric cars?

(A) 30 (B) 34 (C) 35 (D) 32 (E) Other

28. Patrick saw the repeating decimal $0.\overline{34}$ and read it as if it was written in a base other than base 10. He then converted the repeating decimal to a reduced common fraction in base 10. Patrick's result had a denominator and numerator that differed by 13. What is the least possible base that Patrick read the repeating decimal in?

(A) 11 (B) 12 (C) 6 (D) 8 (E) Other

29. What is the sum of the digits of $\sqrt{(2021^2 - 4040)(2021^2 + 4044) - 4}$?

(A) 28 (B) 25 (C) 20 (D) 16 (E) Other

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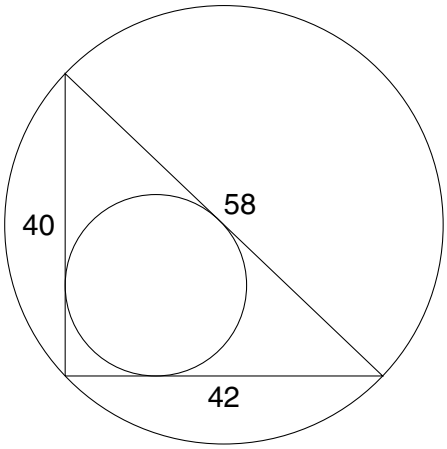
30. The figure below shows a triangle with sides of length 40, 42, and 58, along with an inscribed circle and a circumscribed circle. What is the square of the distance between the centers of the two circles?



(A) 164 (B) 128 (C) 145 (D) 162 (E) Other

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Grade _____

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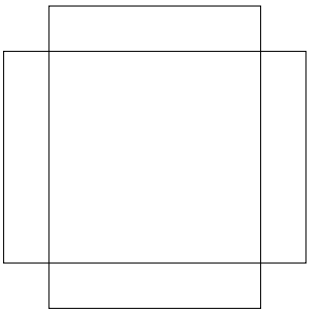
Problems 1 & 2

1. Amy picked a three-digit positive whole number. She gave her friend Claire three clues about the number. Amy said the units digit was twice the value of the hundreds digit, the tens digit was one less than the units digit, and the number was a multiple of three. With this information, Claire knew the number. What was Amy's number?

1.

2. Two rectangles each have a width of 7 and a length of 10. As shown below, these rectangles are placed so that they overlap, and the region common to both rectangles is a square. What is the total area of the region covered by the rectangles?

2.



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Problems 3 & 4

3. Samir summed all of the whole numbers starting with 1 and ending with 15. His result was 101. Samir knew this was incorrect, and realized he did not include two numbers in his sum. One number he did not include was 6. What was the other number?

3.

4. What is the greatest three-digit positive whole number that has a remainder of 2 when divided by 8, a remainder of 3 when divided by 7, and a remainder of 4 when divided by 6?

4.

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School _____

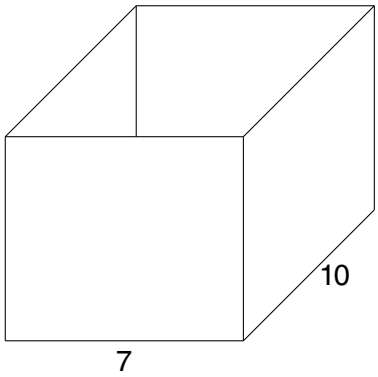
Problems 5 & 6

5. How many two-digit positive whole numbers have a tens digit and units digit that differ by at least three? For example, two such numbers are 94 and 49, since 9 and 4 differ by 5.

5.

6. A box without a lid has a total surface area, inside and outside, of 548. The box has a width of 7 and a depth of 10. What is the height of the box?

6.



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Problems 7 & 8

7. Raymond is playing a game where he scores either five points or seven points on every turn. While playing the game, Raymond scored a total of exactly 99 points. What is the difference between the greatest possible number of turns Raymond could have taken to achieve this score and the least possible number of turns Raymond could have taken to achieve this score?

7.

8. Using each number at most once, Kevin started with 0 and added or subtracted some of the numbers from the list $1, 3, 9, 27$. How many different positive results are possible for Kevin? For example, one such result is $0 + 9 + 27 - 3 = 33$.

8.

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Team Round
12127

School or Team

Name _____

Name _____

Name _____

Name _____

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1.	2.	3.	4.	5.
6.	7.	8.	9.	10.

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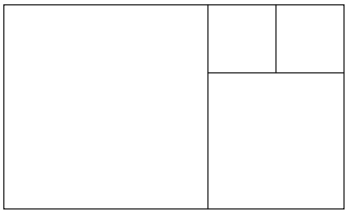
1. What is the sum of the positive whole numbers less than 20 which are not multiples of 2 and are not multiples of 3?

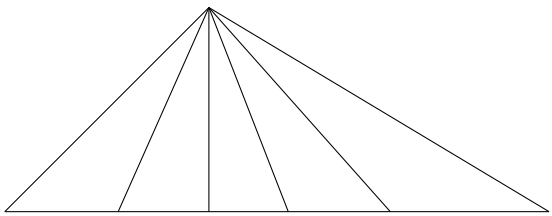
2. A one-day bus and subway ticket costs \$7 for a child, and \$10 for an adult, which is any bus or subway rider age 13 or older. Kevin is 12 years old, and he buys one-day bus and subway tickets for himself and five friends. Two of his friends are 11 years old, one friend is 12 years old, and the rest are older. How many dollars does Kevin pay for all of the one-day bus and subway tickets?

3. The figure below is a rectangle divided into four squares. The two smallest squares each have an area of 4. What is the area of the entire rectangle?



4. Each of the digits 3, 4, 5, 7, and 9 is used exactly once to replace each □ symbol in the expression □□□□ ÷ □. After making the replacements, the expression is a four-digit number divided by a single-digit number. For example, 3457 ÷ 9 is one possibility. If the four-digit number divided by the single-digit number has a result that is a whole number, then what is the greatest possible value of that result?

5. The figure below shows a horizontal line segment and six additional line segments, all with one common endpoint and one endpoint on the horizontal line segment. How many triangles exist in the figure?



6. At recess, Sasha and five friends want to play basketball. How many different ways can Sasha and his friends divide themselves into two teams of three friends each?

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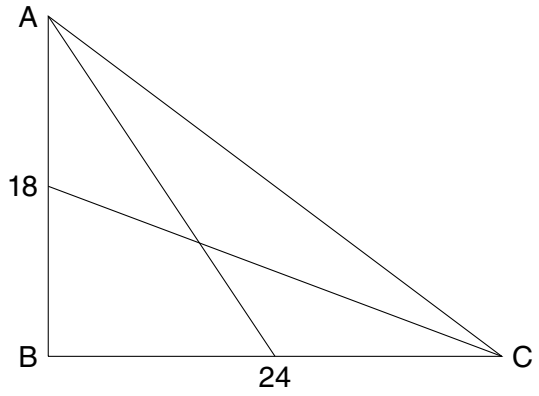
7. Hannah, Isaac, and Jonah worked together cleaning up yards in their neighborhood, and they agreed to split the money they earned. The group used ten dollars of the money they earned to pay back an adult that lent them money to buy supplies. Hannah did the most work, so she received five dollars more than half of what was left after paying back the adult. After the adult and Hannah received their money, Isaac received seven dollars more than one-third of the money that was remaining. Jonah received the final nineteen dollars. Before paying back the adult, how many dollars did Hannah, Isaac, and Jonah earn cleaning up yards?

8. A pebble dropped from a height of $10,000$ feet above the ground fell 16 feet in the first second after it was dropped, and each additional second it fell 32 more feet than it did the second before, so that in the time between the first and second second after it was dropped it fell 48 feet, in the time between the second and third second after it was dropped it fell 80 feet, and so on. What was the average speed of the pebble, in feet per second, from the moment it was dropped until it hit the ground?

9. Right triangle ABC, shown below, has legs of length 18 and 24. A segment from A meets BC at the midpoint of BC, and a segment from C meets AB at the midpoint of AB. These two segments partition ABC into three triangles and a quadrilateral. What is the area of the quadrilateral?



10. Four friends are each given a red chip, a green chip, and a blue chip. Each friend randomly places their three chips into three bags, labeled A, B, and C, so each of the three bags contains exactly one chip from each of the four friends. What is the probability that at least one of the three bags contains four chips that are all the same color? Express your answer as a common fraction.

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Answers 12127

Sprint Round

1. A
2. D
3. D
4. B
5. D
6. E
7. C
8. B
9. C
10. C
11. D
12. D
13. E
14. B
15. A
16. E
17. B
18. A
19. E
20. B
21. C
22. D
23. C
24. E
25. B
26. C
27. B
28. C
29. B
30. C

Target Round

1. 234
2. 91
3. 13
4. 850
5. 49
6. 6
7. 4
8. 40

Team Round

1. 73
2. ($)48
3. 60
4. 1362
5. 15
6. 10
7. ($)98
8. 400
9. 72
10. $\frac{11}{108}$

Number Sense

1. 201
2. 49
3. 345
4. 4
5. 808
6. 131
7. 740
8. 1
9. 12
10. $[1720, 1900]$
11. 957
12. 122
13. 196
14. 8
15. 1170
16. 1040
17. 580
18. 987
19. 1564
20. $[209057, 231061]$
21. 2
22. 2150
23. 6
24. 20
25. 576
26. $\frac{1}{4}$
27. 183
28. $\frac{2}{13}$
29. 34
30. $[112558, 124406]$
31. 7225
32. 13
33. 4257
34. 225
35. 1.5
36. 28
37. 462
38. 5624
39. 42
40. $[5835709, 6449993]$
41. 29593
42. 160
43. 31
44. 2860
45. 16
46. 6561
47. $19\frac{16}{27}$
48. 4
49. 1764
50. $[135170, 149398]$
51. 12390
52. 9
53. 13653
54. 6
55. 47
56. 110111
57. 7896
58. 2080
59. 72
60. $[89, 97]$
61. 55000
62. $\frac{7}{45}$
63. $\frac{1}{221}$
64. $40\frac{16}{81}$
65. 90
66. 11232
67. 41
68. 841
69. 302
70. $[6195, 6846]$
71. 72
72. 16
73. 951600
74. 497025
75. 60
76. 18
77. 128
78. 62100
79. 324
80. $[2146, 2371]$

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Solutions 12127

Sprint Round Solutions

1. This is $(12 \times 14) - (24 \times 6)$, which is $12 \times 14 - 12 \times 12$, or $12 \times 2 = \boxed{24}$.

2. The total number of oranges Jaclyn purchased was $2 \times 12 = 24$. On the first 4 days of the week, she used $4 \times 5 = 20$ oranges, so the number of oranges she had left on Friday was $24 - 20 = \boxed{4}$.

3. This is $567 \times 1000 - 567 \times 1$, which is $567000 - 567$. Since 567 is less than 1000, the thousands digit of the result will be $7 - 1 = \boxed{6}$.

4. The total value of Tamir's quarters is $15 \times 25 = 375$ cents, so the total value of his dimes is $485 - 375 = 110$ cents, so his number of dimes is $110 \div 10 = \boxed{11}$.

5. The smaller square has area $3 \times 3 = 9$. The side length of the larger square is $2 \cdot 3 = 6$, so its area is $6 \times 6 = 36$. The total area of the two squares is $9 + 36 = \boxed{45}$.

6. The greatest three-digit multiple of 27 must be a multiple of 9, and it must be greater than or equal to $999 - 27 + 1 = 973$, since among any 27 consecutive whole numbers there must be at least 1 multiple of 27. The multiples of 9 greater than 973 are 981, 990, and 999. When the number 999 is divided by 9, the result is 111, which is also divisible by 3. Therefore 999 is a multiple of 27. But the number 1 more than 999 is 1000, which is not a three-digit positive whole number. The next greatest positive whole number that is 1 more than a multiple of 27 is $1000 - 27 = \boxed{973}$.

7. The distance Jesse traveled in the first two hours was $2 \cdot 3 = 6$ miles, and the distance Jesse traveled in the final hour was $1 \cdot 2 = 2$ miles. The total distance Jesse traveled, in miles, was $6 + 2 = \boxed{8}$.

8. This is a period of $16 - 1 = 15$ days. There are 24 hours in 1 day, so this is $15 \times 24 = 360$ hours. There are 60 minutes in 1 hour, so the number of minutes is $360 \times 60 = \boxed{21600}$.

9. From each corner of the octagon, diagonals cannot be drawn to the corner or the neighboring corners, so only $8 - 3 = 5$ diagonals can be drawn. Since there are 8 corners, there should be $8 \cdot 5 = 40$ diagonals. But this counts each diagonal twice, since it counts the diagonal at both corners of the shape. Therefore the total number of diagonals is $40 \div 2 = \boxed{20}$.

10. The fractions $\frac{1}{2}$, $\frac{1}{3}$, and $\frac{1}{6}$ sum to 1, and adding in $\frac{1}{1}$, this brings the sum to 2. The fractions $\frac{1}{4}$ and $\frac{1}{5}$ together are a bit less than $\frac{1}{2}$ but more than $\frac{1}{4}$, and the fractions $\frac{1}{7} + \frac{1}{8} + \frac{1}{9}$ also sum to more than $\frac{1}{4}$ but less than $\frac{1}{2}$. Therefore the sum is somewhere between $2\frac{1}{2}$ and 3, and the nearest whole number is $\boxed{3}$.

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11. There are 2 choices for the drivers seat, either Amelia or Daniel. After the driver is chosen, there are 3 choices of the person that sits next to the driver, and the final 2 people can sit in the back of the car in 2 ways. Therefore the total number of ways the four can sit in the car is $2 \times 3 \times 2 = \boxed{12}$.

12. Increasing a number by 25% is multiplying its value by $1 + 0.25 = 1.25$, or $\frac{5}{4}$. Decreasing a number by $16\frac{2}{3}$% is multiplying its value by $1 - 0.1\overline{6} = 0.8\overline{3}$, or $\frac{5}{6}$. Therefore the result is $\frac{5}{6} \cdot \frac{5}{4} \cdot 288 = 25 \cdot 12$, which is $\boxed{300}$.

13. The area of the entire right triangle is $\frac{1}{2} \cdot 30 \cdot 16 = 240$. The grey triangle formed by connecting the midpoints is also a right triangle, with legs that are half the length of the legs of the larger triangle, so its area is $\frac{1}{2} \cdot 15 \cdot 8 = 60$. The area of the three regions of the right triangle that are not gray is $240 - 60 = \boxed{180}$.

14. Since 15 students listed strawberry as their second or least favorite, there were $19 - 15 = 4$ students that listed strawberry as their favorite. Since 11 students listed chocolate as their second or least favorite, there were $19 - 11 = 8$ students that listed chocolate as their favorite. The number of students that listed vanilla as their favorite is $19 - 4 - 8 = \boxed{7}$.

15. The prime factorization of 360 is $2^3 \cdot 3^2 \cdot 5^1$, and the prime factorization of 108 is $2^2 \cdot 3^3$. Any whole number that evenly divides both must also divide the GCD of 360 and 108, which is $2^2 \cdot 3^2 = 36$. The number of positive whole number divisors of 36, can be found by taking the product of 1 more than the value of each exponent in its prime factorization, $2^2 \cdot 3^2$, so the result is $(2 + 1)(2 + 1) = \boxed{9}$.

16. Consider the number of chocolate chip cookies on a tray. If there are 4 chocolate chip cookies, there is only 1 possible tray, as there must be exactly 1 of both the peanut butter and sugar cookies. If there are 3 chocolate chip cookies, there are 2 possible trays, as there could be either 1 or 2 peanut butter cookies, with the remainder being sugar cookies. Similarly, if there are 2 chocolate chip cookies, there are 3 possible trays, and if there is only 1 chocolate chip cookie, there are 4 possible trays. Altogether, the total number of possible trays is $4 + 3 + 2 + 1 = \boxed{10}$.

17. By the Pythagorean Theorem, the length of one diagonal is $\sqrt{9^2 + 7^2} = \sqrt{130}$, so the sum of the length of both diagonals is $2\sqrt{130}$, or $\sqrt{4} \cdot \sqrt{130} = \sqrt{520}$. Since $22^2 = 484$ and $23^2 = 529$, the nearest whole number is $\boxed{23}$.

18. While the average could be calculated directly, the ratio of $135 : 189$ simplifies to $5 : 7$. Then the average is $(5 \cdot 52 + 7 \cdot 64) \div (5 + 7)$, or $708 \div 12 = \boxed{59}$.

19. Observe that $10000 - 2021 = 7979$, $100000 - 2021 = 97979$, $1000000 - 2021 = 997979$, and $10000000 - 2021 = 9997979$. In general, when 2021 is subtracted from a number that is 1 followed by 4 or more zeros, the result has a number of digits equal to the number of zeros, with two digits having the value 7 and the rest of the digits having the value 9. Therefore the sum of the digits of the result is $2019 \cdot 9 + 2 \cdot 7 = \boxed{18185}$.

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20. Number the seats from 1 through 50. Seats 1 and 2 cannot both be unoccupied, or else Janice could sit in seat 1 and not have anyone sitting next to her. If seat 1 is unoccupied, then seat 2 must be occupied. Seats 3 and 4 can both be unoccupied, but seat 5 must be occupied. This pattern continues, with seats $2, 5, 8, \ldots, 50$ all occupied, for a total of $\frac{50-2}{3} + 1 = 17$ occupied seats. If seat 1 is occupied, then seats 2 and 3 can both be unoccupied, but seat 4 must be occupied. This pattern continues, with seats $1, 4, 7, \ldots, 49$ all occupied, for a total of $\frac{49-1}{3} + 1 = 17$ occupied seats. The fewest number of seats that could have been sitting in the row when Janice was choosing her seat was 17.

21. The given expression can be rewritten as $a^2 + 2ab + b^2 = (a + b)^2$, so $91\heartsuit 8 = (91 + 8)^2$, and $41\heartsuit 8 = (41 + 8)^2$. Finally, $(91 + 8)^2 - (41 + 8)^2$ is $99^2 - 49^2$. By difference of squares, this is $(99 + 49)(99 - 49) = 148 \cdot 50$, or $\boxed{7400}$.

22. The two triangles are isosceles right triangles with hypotenuses of length 2 and 5, respectively, and legs of length $\frac{5}{\sqrt{2}}$ and $\frac{2}{\sqrt{2}}$. These legs are sides of the rectangle, so the area of the rectangle is $\frac{5}{\sqrt{2}} \cdot \frac{2}{\sqrt{2}} = \frac{10}{2}$, which is $\boxed{5}$.

23. If $5^{2-x} = 12.5$, then $5^2 \cdot 5^{-x} = 12.5$. Dividing both sides by 5^2 yields $5^{-x} = \frac{1}{2}$, so $5^x = 2$. Then $(5^x)^3 = 2^3$, or $125^x = 8$. Finally $125^{x+1} = 8 \cdot 125$, which is $\boxed{1000}$.

24. There were $154 - 68 = 86$ students wearing jeans, and $68 - 29 \cdot 2 = 10$ students wearing shorts paired with a student wearing jeans. So $86 - 10 = 76$ students wearing jeans paired with another student wearing jeans. The number of pairs of students where both students wore jeans was $\frac{76}{2} = \boxed{38}$.

25. Of the 52 cards in the deck, 12 are jacks, queens, or kings, and 40 are not. There are $\binom{40}{1} \cdot \binom{12}{1} = 480$ possible sets of two cards where exactly one of the two cards is a jack queen or king. The total number of sets of two cards is $\binom{52}{2} = 51 \cdot 26$, for a probability of $\frac{480}{51 \cdot 26} = \frac{80}{17 \cdot 13}$. This is $\frac{80}{221}$, and $80 + 221 = \boxed{301}$.

26. The prime factorization of 288 is $2^5 \cdot 3^2$. The sum of the divisors of 288 that are multiples of 3 is $2^0 \cdot 3^1 + 2^1 \cdot 3^1 + \cdots + 2^5 \cdot 3^1 + 2^0 \cdot 3^2 + 2^1 \cdot 3^2 + \cdots + 2^5 \cdot 3^2$, or $(3^1 + 3^2)(2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5)$. The sum of the divisors of 288 that are not multiples of 3 is $2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5$. The first sum divided by the second sum is $3^1 + 3^2 = \boxed{12}$.

27. Since 3 workers can assemble 5 cars in 30 days, each car requires $\frac{3 \cdot 30}{5} = 18$ worker-days to complete, so the 7 cars being assembled will require $18 \cdot 7 = 126$ worker-days. Including the 5 days at the start of assembly and the 7 days at the end of assembly, the 5 workers assembling the cars contribute $5 \cdot (5 + 7) = 60$ worker-days. The remaining $126 - 60 = 66$ worker-days take the 3 workers $\frac{66}{3} = 22$ days to complete. From start to finish, the number of days required for the assembly process is $12 + 22 = \boxed{34}$.

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28. Let the base be b and the value of the repeating decimal be v, so that $v = 0.\overline{34}_b$. Then $v \cdot b^2 = 34.\overline{34}_b$, and $v \cdot b^2 - v = 34_b$. The right side can be expressed in base 10, yielding $v \cdot (b^2 - 1) = 3b + 4$, so $v = \frac{3b+4}{b^2-1}$. The value v must be less than 1 and is equal to a fraction with a numerator that is a multiple of 13 less than the denominator, so $b^2 - 1 - 13k = 3b + 4$, and $b^2 - 3b - (13k - 5) = 0$. For positive values of k, this equation has a positive and negative root with absolute values that differ by 3, and the value of the positive root increases as k increases. Trying different values of k, when $k = 1$, the quadratic $b^2 - 3b - 18$ can be factored as $(b - 6)(b + 3)$, so the least possible value for b is $\boxed{6}$.

29. Let $x = 2021$. Then the expression under the radical becomes $(x^2 - 2x + 2)(x^2 + 2x + 2) - 4$. Taking the product of the two trinomials yields $(x^4 + 4) - 4$, or x^4. The square root of x^4 is x^2, and $2021^2 = 4084441$. The sum of the digits of this number is $\boxed{25}$.

30. The triangle is similar to a triangle with sides of length 20, 21, and 29, which is a Pythagorean triple, so the triangle is a right triangle, and the center of the circumscribed circle is at the midpoint of the hypotenuse. Additionally, the radius of the inscribed circle is equal to the area of the triangle divided by the semiperimeter of the triangle, or $\frac{\frac{1}{2} \cdot 42 \cdot 40}{\frac{1}{2} \cdot (40+42+58)} = 12$. Now impose a coordinate system on the triangle, with the vertex that is the right angle at $(0, 0)$. The center of the inscribed circle is at $(12, 12)$, and the midpoint of the hypotenuse is $(21, 20)$. The square of the distance between $(12, 12)$ and $(21, 20)$ is $(21 - 12)^2 + (20 - 12)^2 = \boxed{145}$.

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Target Round Solutions

1. With the units digit being twice the value of the hundreds digit and the tens digit being 1 less than the units digit, the only possibilities for the number are 112, 234, 356, and 478. Of these, only one is a multiple of 3, and it is $\boxed{234}$.

2. The total area of the two rectangles is $2 \cdot 7 \cdot 10 = 140$, but this counts twice the area of the square where the rectangles overlap. The area of this square is $7 \cdot 7 = 49$, so the total area covered by the two rectangles is $140 - 49 = \boxed{91}$.

3. The sum of the whole numbers from 1 through 15 is $\frac{1}{2} \cdot 15 \cdot (15 + 1) = 120$, so the sum of the two numbers not included was $120 - 101 = 19$. Since one of the numbers not included was 6, the other number not included was $19 - 6 = \boxed{13}$.

4. Considering numbers that are 3 more than multiples of 7, the number 10 is also both 2 more than a multiple of 8 and 4 more than a multiple of 6. Any other number that has the same remainder when divided by 7, 8, and 6 must differ from 10 by a multiple of 7, 8, and 6. The least common multiple of 7, 8, and 6 is $7 \cdot 3 \cdot 2^3 = 168$. The greatest multiple of 168 that is less than 1000 is $5 \cdot 168 = 840$, and $10 + 840 = \boxed{850}$.

5. If the tens digit is 1, the units digit can be 4 through 9, for 6 possible numbers. If the tens digit is 2, the units digit can be 5 through 9, for 5 possible numbers. For tens digits of 3 through 7, each increase in the tens digit removes 1 possibility from the upper range of the units digit, but adds 1 possibility in the lower range of the units digit, so the number of possible values of the units digit in each case stays constant at 5. So for tens digits of 2 through 7, there are $6 \cdot 5 = 30$ such numbers. If the tens digit is an 8, there are 6 possible values for the units digit, and if the tens digit is 9, there are 7 possible values for the units digit. Altogether, the total is $6 + 30 + 6 + 7 = \boxed{49}$.

6. The total surface area of the bottom of the box, inside and outside, is $7 \cdot 10 \cdot 2 = 140$, so the remaining surface area of the box is $548 - 140 = 408$. Let the height of the box be h. The remaining surface area is comprised of 4 rectangles with area $7h$, and another 4 rectangles with area $10h$, for a total area of $4 \cdot 7h + 4 \cdot 10h = 68h$. Since $68h = 408$, the height of the box is $\frac{408}{68} = \boxed{6}$.

7. Let F be the number of 5-point turns and S be the number of 7-point turns, so $5F + 7S = 99$. To maximize the number of turns needed, the number of 5-point turns must be maximized. Subtracting increments of 7 from 99 until a multiple of 5 is encountered, this occurs at $5 \cdot 17 + 7 \cdot 2 = 99$, for a maximum number of turns of $17 + 2 = 19$. To minimize the number of turns needed, the number of 7-poin turns must be maximized. Subtracting increments of 5 from 99 until a multiple of 7 is encountered, this occurs at $5 \cdot 3 + 7 \cdot 12 = 99$, for a minimum number of turns of $3 + 12 = 15$. The difference between the greatest and least possible number of turns is $19 - 15 = \boxed{4}$.

8. Every possible sum can be expressed as $1a + 3b + 9c + 27d$, where a, b, c, or d are each -1, 0, or 1. If $1 + 3 + 9 + 27 = 40$ is added to each of these results, then a, b, c, and d are all 0, 1, or 2. This is equivalent to all possible positive base 3 numbers that can be written with 4 digits. There are $3^4 = 81$ such numbers, ranging from 0 through 81. When 40 is subtracted from each of these values, the number of positive values is $81 - 40 - 1 = \boxed{40}$.

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Team Round Solutions

1. The numbers that are not multiples of 2 and are not multiples of 3 are odd numbers that are not divisible by 3, and their sum is $1 + 5 + 7 + 11 + 13 + 17 + 19 = \boxed{73}$.

2. Kevin buys a total of $1 + 5 = 6$ tickets. The tickets for himself, his two 11-year-old friends, and his one 12-year-old friend are \$7 each and cost $4 \times \$7 = \$28$ altogether. The tickets for the $5 - 2 - 1 = 2$ friends older than age 12 are \$10 each, and cost $2 \times \$10 = \$20$ altogether. The total amount Kevin spends for subway tickets is $\$28 + \$20 = \boxed{\$48}$.

3. The side length of one of the small squares is $\sqrt{4} = 2$. The side length of the medium square is therefore $2 + 2 = 4$, and the side length of the largest square is $2 + 4 = 6$. The rectangle has a width of $6 + 2 + 2 = 10$, a height of $4 + 2 = 6$, and an area of $10 \times 6 = \boxed{60}$.

4. The single-digit number cannot be 5, as the four-digit number cannot be a multiple of 5. The single-digit number cannot be 4, as the four-digit number cannot be even, and therefore cannot be a multiple of 4. The single-digit number cannot be 3, as the sum of the other four digits, $5 + 4 + 7 + 9 = 25$, is not a multiple of 3. Similarly, the single-digit number cannot be 9, as the sum of the other four digits, $5 + 4 + 7 + 3 = 19$, is not a multiple of 9. Therefore the single-digit number must be 7. To maximize the result, the four-digit number must be as large as possible and a multiple of 7. Evaluating possibilities, the greatest possible four-digit number, 9543, is not a multiple of 7, but the second greatest possible four-digit number, 9534 is a multiple of 7, and $9534 \div 7 = \boxed{1362}$.

5. Any triangle must have two sides that are the non-horizontal segment, and one side that is a portion of the horizontal segment. The horizontal segment is partitioned into 5 sub-segments by the non-horizontal segments. There is 1 triangle that uses the entire horizontal segment. There are 2 triangles that use 4 adjacent sub-segments. There are 3 triangles that use 3 adjacent sub-segments. There are 4 triangles that use 2 adjacent sub-segments. There are 5 triangles that use exactly 1 sub-segment. Altogether, the total number of triangles is $1 + 2 + 3 + 4 + 5 = \boxed{15}$.

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6. Consider the team that has Sasha. There are $\binom{5}{2} = 10$ different pairs of teammates for Sasha, and the remaining three friends must be on the other team. Therefore the total number of ways to form the two teams is $\boxed{10}$.

7. If Isaac received 7 dollars more than $\frac{1}{3}$ of what was remaining when he was paid, then Jonah received 7 dollars less than $\frac{2}{3}$ of what was remaining when Isaac was paid. Then $\frac{2}{3}$ of what was remaining was $7 + 19 = 26$, and the amount remaining before Isaac and Jonah were paid was $\frac{3}{2} \cdot 26 = 39$ dollars. Since Hannah was paid 5 dollars more than half of what remained after paying back the adult, the 39 dollars is 5 less than half of what remained. The amount remaining after paying back the adult was $(39 + 5) \cdot 2 = 88$ dollars, and the amount earned before paying back the adult was $\$88 + \$10 = \boxed{\$98}$.

8. The distances the pebble travels each second after it was dropped are the arithmetic sequence $16, 48, 80, 112, \ldots$, so that n seconds after it was dropped, the pebble fell $16 + 32(n - 1)$ feet in the preceding second. The sum of the arithmetic series $16 + 48 + 80 + 112 + (16 + 32(n - 1))$ is $10,000$ when the pebble hit the ground. This series has n terms, with an average value of $\frac{1}{2}(16 + (16 + 32(n - 1))) = 16n$, and therefore a sum of $n \cdot 16n = 16n^2$. When $16n^2 = 10,000$, the value of n is 25, so the pebble hit the ground 25 seconds after it was dropped. The average speed of the pebble, in feet per second, is $10,000$ feet divided by 25 seconds, which is $\boxed{400}$.

9. The area of triangle ABC is $\frac{1}{2} \cdot 18 \cdot 24 = 216$. Let the point where the segment from A to BC meets BC be X, and the point where the segment from C to AB meets AB be Y. Since triangle ABX and AXC share an altitude from A and the length of segment BX is equal to the length of segment XC, the area of triangle AXC is $\frac{1}{2} \cdot 216 = 108$. Additionally, segment AX and segment CY are medians of the triangle. Let the point where the two segments intersect be Z. Then the length of ZX is $\frac{1}{3}$ the length of AX, and the area of triangle ZXC is $\frac{1}{3}$ the area of triangle AXC, or $\frac{1}{3} \cdot 108 = 36$. As with AXC, the area of triangle CBY is $\frac{1}{2} \cdot 216 = 108$. Therefore the area of quadrilateral $AXZY$ is $108 - 36 = \boxed{72}$.

10. There are $3! = 6$ ways for each of the 4 friends to place their chips into the bags, for a total number of possibilities of $6^4 = 1296$. For exactly one of the bags to contain chips of the same color, there are $\binom{3}{1} = 3$ ways to select the bag that contains the identical chips, $\binom{3}{1} = 3$ ways to select the color of the chip that is in the bag, and $2^4 = 16$ ways for each of the four friends to place their remaining chips in the other two bags. However, 2 of these 16 arrangements result in all three of the bags containing chips of the same color. Thus the number of ways to have exactly one bag contain chips of the same color is $3 \cdot 3 \cdot (16 - 2) = 126$. If two bags contain chips that are the same color, then all three bags contain chips that are the same color, which can be done in 6 ways, choosing the colors each bag holds. Therefore the total number of ways for at least one bag to contain 4 chips of the same color is $126 + 6 = 132$, for a probability of $\frac{132}{1296} = \boxed{\frac{11}{108}}$.

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